Indoor Location Prediction from Wi-Fi Signal Strength: A Machine Learning Baseline on UJIIndoorLoc

Rishi Raj (2401010380), Aditya Samadhiya (2401010037), Yash Yadav (2401010515),
Kaustubh Ranjan Mishra (2401020116), Yash Raj (2401010514)
Advanced Machine Learning, Project 9, Phase 1
Section D, Rishihood University

Abstract—GPS does not work reliably indoors, but almost every building is covered by Wi-Fi. We predict a phone's building, floor and (x, y) position from a single scan of Wi-Fi received signal strengths (RSSI), using the public UJIIndoorLoc dataset (3 buildings, 13 building–floor classes, 520 access points, 21,048 scans). The data is extremely sparse: 96.5% of the readings are "not detected". We clean the data, handle the missing-signal convention consistently, and compare linear/logistic regression, k-nearest neighbours and Random Forests, including a hierarchical Random Forest that first predicts the building and then uses a per-building model. All choices are made by 5-fold cross-validation grouped by user, and the official validation file, collected four months later, is used only once as the test set. On the test set, the per-building Random Forest reaches 99.8% building accuracy, 91.3% floor accuracy and 9.37 m mean position error (7.07 m median). k-NN has the lowest mean error (8.69 m). Linear regression does badly at positioning (52 m), which shows that the mapping from signal to position is strongly non-linear. We also analyse the errors by building/floor correctness, by phone and by signal coverage, and point out what Phase 2 (multitask deep learning) should improve.

Index Terms—indoor positioning, Wi-Fi fingerprinting, RSSI, random forest, k-nearest neighbours, UJIIndoorLoc

I. Introduction

Satellite positioning fails inside buildings because walls and roofs block and reflect the signal. Indoor location is still useful for navigation in campuses and hospitals, for emergency response and for asset tracking. Wi-Fi access points (APs) are already everywhere, and a phone can measure the RSSI of every AP it hears without connecting to it. Converting RSSI to distance with a propagation model is unreliable indoors because of walls, people and multipath reflections [1]. Fingerprinting avoids modelling propagation: a radio map of RSSI vectors is recorded at known positions, and a new scan is located by learning from that map [2].

The task combines two classification problems (building, floor) and one regression problem (x, y in metres). Phase 1 of this project builds and validates the machine learning baseline. Our contributions are:

- a careful data audit (duplicates, empty scans, a faulty phone, train/test shift);
- a leakage-free pipeline in which every statistic is fitted on training data, and model selection uses cross-validation that holds out whole users;
- a comparison of linear/logistic regression, k-NN and Random Forests (plain and hierarchical) on building accuracy, floor accuracy, coordinate error and the IPIN competition score;
- an error analysis with a building/floor breakdown, floor confusion matrices, coordinate error maps and errors per phone.

II. Literature Review

Fingerprinting. RADAR [2] introduced Wi-Fi fingerprinting with nearest-neighbour matching in signal space. Horus [3] modelled the RSSI distribution at each location probabilistically, which showed how noisy single readings are.

UJIIndoorLoc. Torres-Sospedra et al. [4] released our dataset, the first large multi-building, multi-floor public benchmark, with a k-NN baseline reaching 89.9% floor accuracy and 7.9 m mean error. A follow-up study [5] compared 51 distance measures and four data representations for k-NN and found that the powed representation, which compresses weak readings, helps. The IPIN 2016 competition [6] defined the combined score we report.

Machine learning models. Bozkurt et al. [7] compared decision trees, naive Bayes, k-NN and ensembles on UJIIndoorLoc and found k-NN and tree ensembles strongest. HybLoc [8] splits the radio map into clusters and trains a Random Forest [9] per cluster, the same divide-and-conquer idea as our per-building model.

Deep learning (Phase 2 context). Stacked autoencoders with an MLP [10], [11] and CNNs [12] reach about 92–96% floor accuracy on UJIIndoorLoc. Multitask learning [13], where related tasks share one representation, motivates our Phase 2 model.

Gaps addressed here. (i) The "+100 = not detected" value is often fed to models as a number, which inverts its meaning. (ii) Random train/validation splits leak near-duplicate scans and overstate accuracy. (iii) Most baselines model building, floor and position independently.

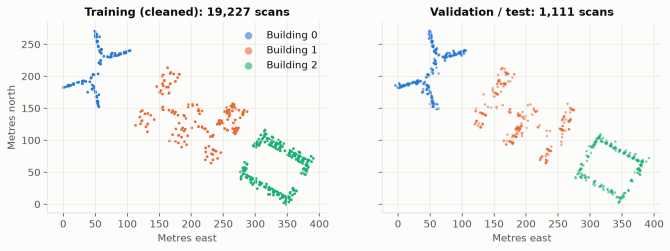


Fig. 1. Scan positions. The test set covers the same buildings but is skewed towards building 0.

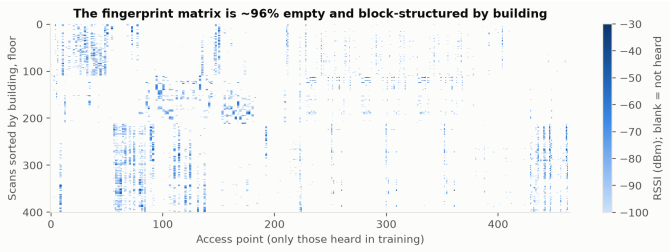


Fig. 2. Sample of the fingerprint matrix (rows sorted by building and floor). It is about 96% empty, with block structure per building.

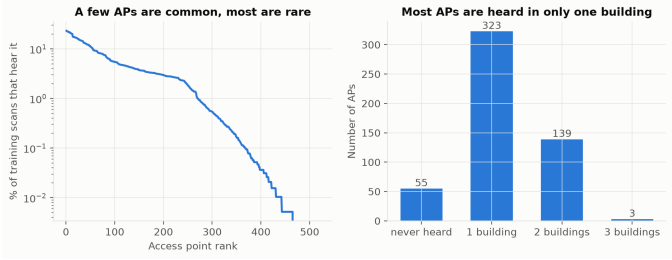


Fig. 3. Left: a few APs are heard often, most rarely. Right: most APs are heard in a single building.

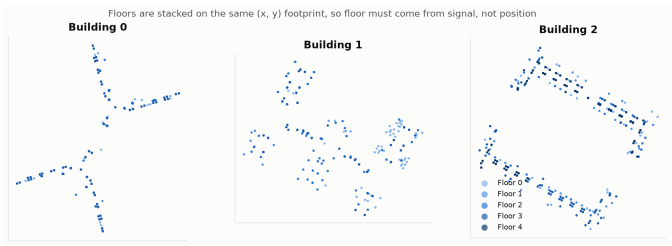


Fig. 4. Training positions coloured by floor. Floors are stacked on the same (x, y) footprint.

III. Dataset and Exploratory Analysis

UJIIndoorLoc [4] was collected on the Universitat Jaume I campus: three buildings with four, four and five floors. Each scan contains 520 RSSI values (dBm; +100 means not detected) plus longitude and latitude in metres (UTM), floor, building, space ID, user ID, phone ID and a timestamp. The training file has 19,937 scans from 18 users and 16 phones (May–June 2013). The validation file has 1,111 scans collected four months later (Sept–Oct 2013) by different users, on 11 phones, 9 of which do not appear in training.

Cleaning. We removed 637 exact duplicate rows and 73 scans in which no AP was heard, leaving 19,227 training scans. The test file is left untouched so that our results stay comparable with the literature.

Sparsity. A scan hears about 18 of the 520 APs on average, so 96.5% of the matrix is "not detected" (Fig. 2). Detected RSSI ranges from −104 to 0 dBm, and most readings are weaker than −70 dBm. 55 APs are never heard in training, and 323 of the remaining 465 are heard in only one building (Fig. 3). This is why building classification is easy.

Floors. Floors share the same footprint (Fig. 4), so position cannot reveal the floor; it must come from which APs are heard and how strongly.

Class balance and shift. Training is fairly balanced over the 13 building–floor classes. The test set is skewed: 48% of test scans are in building 0, compared with 27% in training (Fig. 5). Test phone proportions also differ (Fig. 6), and 64% of test scans come from phones never seen in training. So the test set checks how well the models work for new users and new phones at a later date.

Data quality. 492 training scans contain readings above −30 dBm, which is implausibly strong for indoor Wi-Fi. 430 of them come from user 6 with phone 19 (Fig. 7). We keep them by default and test the effect of removing them in Section V.

PCA. With the missing values handled, the first two principal components already separate the three buildings (Fig. 8), but 50 components explain only 84% of the variance. The useful information is spread over many APs.

IV. Methodology

A. Problem formulation

Let $\mathbf{x} \in \mathbb{R}^{520}$ be a scan. We learn $f_b(\mathbf{x}) \in \{0, 1, 2\}$, $f_f(\mathbf{x}) \in \{0, \ldots, 4\}$ and $f_p(\mathbf{x}) \in \mathbb{R}^2$. Following the IPIN competition [6], the combined score of one prediction is

$$s = \|\mathbf{p} - \hat{\mathbf{p}}\|_2 + 50 \cdot [b \neq \hat{b}] + 4 \cdot [f \neq \hat{f}], \qquad (1)$$

so a wrong building costs as much as 50 m of error and a wrong floor costs 4 m.

B. Feature engineering

Missing signals. "Not detected" means the signal is below the phone's sensitivity, so it is missing not at random: its absence tells us the AP is far away. Used as a number, +100 dBm would look like the strongest possible signal. We replace it with $m = \min(\text{observed}) - 1 = -105$ dBm, computed on training data only. APs never heard in training are dropped, which leaves 465 features.

Variants compared. (a) raw: dBm with −105 for not detected; (b) positive: $x - m$, so not detected is 0; (c) powed: $\big((x - m)/(x_{\max} - m)\big)^e$, which compresses weak readings [5]; (d) raw plus one binary heard/not-heard flag

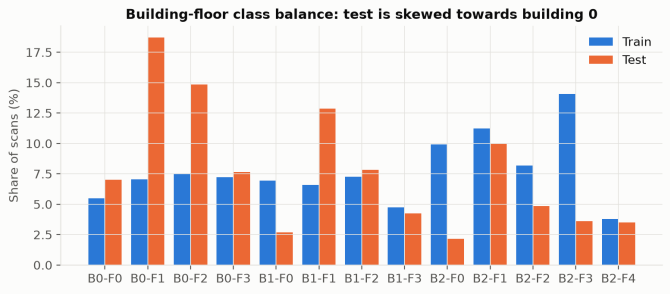


Fig. 5. Share of scans per building–floor in train and test.

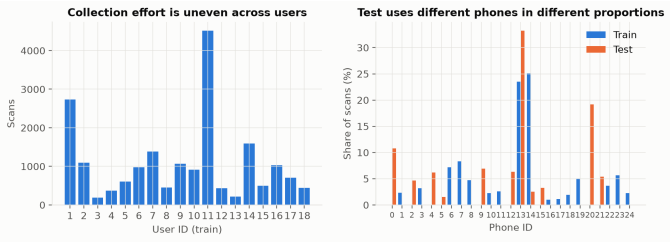


Fig. 6. Scans per user (train) and per phone (train vs. test).

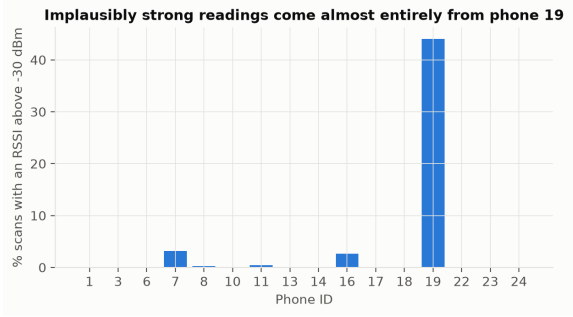


Fig. 7. Share of scans with an RSSI above −30 dBm, by phone.

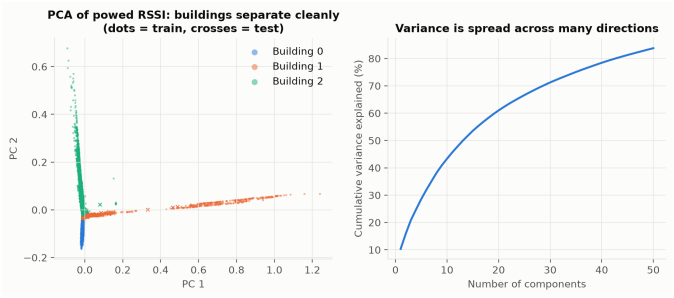


Fig. 8. PCA of the RSSI matrix. Buildings separate; variance is spread across many directions.

per AP; (e) (d) plus per-scan summaries (number of APs heard, strongest, mean and standard deviation of RSSI, and the IDs of the three strongest APs). The variant for each model is chosen by cross-validation.

C. Models

Centroid baseline. Predicts the most common building and floor and the mean position. Any useful model should do better than this.

Linear and logistic regression. Inputs are standardised with training statistics. Logistic regression (softmax, cross-entropy, L2, $C = 1$) predicts building and floor. Ordinary least-squares linear regression predicts x and y.

k-NN [14]. The classic fingerprinting method: find the k training scans nearest in Euclidean distance. Building and floor come from a distance-weighted vote, and the position is the distance-weighted mean of the neighbours' positions.

Random Forest [9]. An ensemble of CART trees. Each tree is grown on a bootstrap sample, and each split considers a random subset of max_features APs. Splits minimise Gini impurity $G = 1 - \sum_k p_k^2$ for classification and variance for regression. For B trees with variance σ^2 and pairwise correlation ρ, the variance of their average is

$$\mathrm{Var}\left(\frac{1}{B}\sum_b T_b\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2, \qquad (2)$$

so bagging removes the second term as B grows and random feature selection lowers ρ. Because a tree only compares a feature with a threshold, any order-preserving transform of the input gives the same tree. A regression leaf predicts the mean of real training positions, so predictions never leave the surveyed area.

Plain vs. per-building (hierarchical). The plain variant trains three independent forests on all scans, as in the project brief. The per-building variant predicts the building first, then sends the scan to a floor forest and a position forest trained only on that building's scans. This splits one large problem into three smaller and more similar ones. The downside is that a wrong building prediction also makes the floor and position wrong.

D. Validation protocol

Consecutive scans by the same user at the same spot are near-duplicates, so a random split would leak them. We use 5-fold GroupKFold with USERID as the group: no user appears in both the training and validation parts of a fold. Feature variants, k and max_features are all selected by the mean CV IPIN score. Imputation and scaling are re-fitted inside every fold. The official validation file is used once, as the test set, after all choices are fixed.

V. Results

A. Feature comparison

Table I compares the feature variants. The raw and positive variants give identical results for both models, as theory predicts: the shift is order-preserving (trees) and cancels in Euclidean distances (k-NN). None of the richer variants improved the CV score, so the simplest representation (raw dBm with −105 for not detected) is used for every model. Powed features helped k-NN on the test set (IPIN 8.58 vs. 9.22) but hurt it in CV (17.60 vs.

TABLE I
Feature comparison. CV = 5-fold GroupKFold by user on the training set (used to choose); test shown for reference only.

Features	CV IPIN	Test IPIN	Test B+F %	Test err. (m)
Random Forest (per-building)				
raw dBm, not-heard = -105	15.23	9.70	91.18	9.31
positive (shifted so not-heard = 0)	15.23	9.70	91.18	9.31
powed (compress weak signals)	15.71	9.88	91.09	9.44
raw + heard/not-heard flags	15.55	10.28	90.28	9.80
raw + flags + scan summary	15.82	9.86	91.00	9.50
k-NN (k=5)				
raw dBm, not-heard = -105	14.46	9.22	90.10	8.69
positive (shifted so not-heard = 0)	14.46	9.22	90.10	8.69
powed (compress weak signals)	17.60	8.58	93.43	8.19
raw + heard/not-heard flags	14.54	9.20	90.01	8.66

TABLE II
Hyper-parameter search with user-grouped 5-fold CV on the training set.

Model	Parameters	B+F %	Mean err. (m)	IPIN
k-NN	k=1	72.38	12.92	15.55
k-NN	k=3	72.30	12.10	14.71
k-NN	k=5	72.29	11.78	14.46
k-NN	k=7	72.26	11.80	14.73
k-NN	k=11	72.06	11.93	15.05
Random Forest	max_features=sqrt, n_estimators=300	70.90	13.55	15.21
Random Forest	max_features=0.1, n_estimators=300	71.56	14.38	16.08
Random Forest	max_features=0.3, n_estimators=300	70.93	15.57	17.45

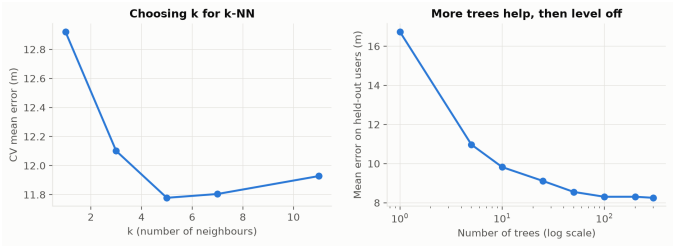


Fig. 9. Left: CV error vs. k. Right: error on held-out users vs. number of trees.

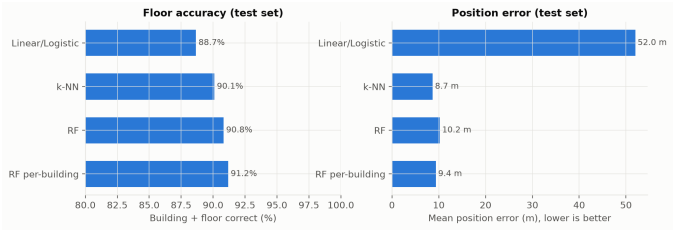


Fig. 10. Floor accuracy and mean position error on the test set (centroid baseline omitted: 3.6%, 142.8 m).

14.46). Because test scores must not drive selection, we did not adopt them.

B. Tuning

Table II shows the CV search. For k-NN, $k = 5$ is best: $k = 1$ is noisy and $k = 11$ over-smooths. For the Random Forest, max_features= $\sqrt{p}$ (≈21 APs per split) beats 10% and 30%, because more random, less correlated trees average better. The error drops quickly up to about 50 trees and is flat beyond 100 (Fig. 9), so 300 trees is safely on the plateau. In CV, k-NN scores slightly better than the forest (IPIN 14.46 vs. 15.21; floor accuracy 74.1% vs. 71.3%), while the forest identifies the building more reliably (99.0% vs. 96.7%).

C. Test-set comparison

Table III and Fig. 10 report the single evaluation on the official test set. All learned models identify the building almost perfectly. Logistic regression even gets 100%, because buildings use disjoint APs. On floor, the per-building Random Forest is best (91.3%), ahead of the plain forest (90.9%), k-NN (90.1%) and logistic regression (88.7%). For position, linear regression is unusable (52.0 m), while k-NN (8.69 m) and the per-building forest (9.37 m) are close. The per-building design lowers the forest's mean error from 10.22 m to 9.37 m and its 95th-percentile error from 27.4 m to 23.3 m, which is the lowest of all models (Fig. 11). For reference, the dataset paper's k-NN baseline reported 89.9% floor accuracy and 7.9 m mean error [4].

D. Error analysis

Hierarchical breakdown (Table IV). For the per-building forest, only 2 of 1,111 scans have the wrong building, but they are far off (107 m). The 96 scans with the right building but wrong floor average 15.7 m, almost twice the 8.6 m of fully correct scans. Most of the total error (83%) still comes from correct-floor scans, so improving within-floor positioning matters as much as floor classification.

Floors (Fig. 12). Building 0 is easy (97.6%). Building 1 is the weak spot (79.8%): 43 of its 143 floor-1 test scans are predicted as floor 2. Errors are almost always between adjacent floors, as expected when signals leak through one ceiling.

Spatial errors (Fig. 13). Building 0 has the lowest error (6.5–7.5 m per floor). The worst building–floors are building 2 floor 2 (17.6 m) and floor 4 (16.2 m).

Coverage and devices (Figs. 14, 15). Scans that hear 5 or fewer APs have the largest error (27.9 m, $n = 18$). Error also rises for scans that hear more than 25 APs. Of these 141 scans, 105 come from phone 20, which is not in the training data. This phone hears more APs per scan than any other test phone (25 on average), and it is one of the least accurate phones (13.2 m). Across test phones, mean error ranges from 6.2 m (phone 9) to 16.3 m (phone 14), so differences between phones clearly add to the error.

Outliers and importance. Removing the 492 scans with readings above −30 dBm did not help (9.46 m vs. 9.37 m), so these readings do not seem to hurt the forest. Feature importance (Fig. 16) shows that the building classifier spreads its importance over many APs instead of relying on a single one.

VI. Discussion

Why linear models fail at position but not at building. Building identity is almost a linear function of which APs are heard, so a weighted sum separates it. Position is not: RSSI falls with the logarithm of distance, walls cut it abruptly, and the same AP is "east of you" in one corridor

TABLE III
Test-set results (official UJIIndoorLoc validation file, 1,111 scans). Accuracies in %, errors in metres. MAE lon./lat. = mean absolute error of each coordinate. IPIN score = 2-D error + 50× building miss + 4× floor miss.

Model	Building	Floor	Bldg+Floor	MAE lon.	MAE lat.	Mean err.	Median	P75	IPIN score
Centroid (guess the average)	24.12	15.48	3.60	122.42	66.20	142.75	167.37	193.37	184.07
Linear + Logistic Regression	100.00	88.66	88.66	43.70	24.67	51.99	37.53	72.25	52.45
k-NN	99.73	90.10	90.10	5.80	5.35	8.69	5.35	10.31	9.22
Random Forest	99.82	90.91	90.82	6.82	6.20	10.22	7.63	13.35	10.68
Random Forest (per-building)	99.82	91.27	91.18	6.19	5.77	9.37	7.07	12.19	9.81

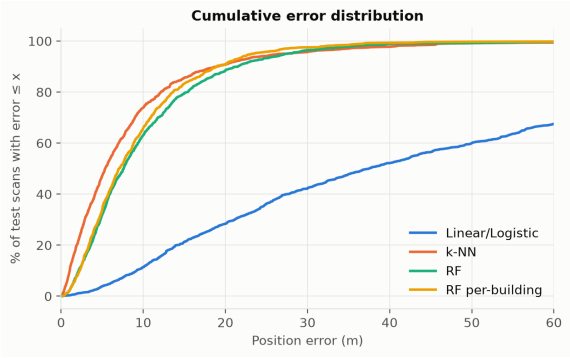


Fig. 11. Cumulative distribution of position error.

TABLE IV
Error split by whether building and floor were right, Random Forest (per-building) (test set).

Case	n	Mean (m)	Median (m)	Share of total error %
building ok, floor ok	1013	8.57	6.65	83.4
building ok, floor wrong	96	15.73	12.55	14.5
building wrong	2	107.25	107.25	2.1

and "west" in another. A single hyperplane cannot model this, but trees and k-NN make local, piecewise decisions, so they can.

k-NN vs. Random Forest. The two are close, and which one wins depends on the evaluation. k-NN is ahead in CV. On the test set it has the lower mean (8.69 m) and median (5.36 m) error, while the per-building forest has the best floor accuracy and the smallest large errors. k-NN copies the positions of real neighbouring fingerprints, which works well when the radio map is dense. The forest learns which APs matter. We keep the Random Forest as the main model because the brief asks for it and it is best on floor, which is the hardest target on the test set. We report k-NN next to it.

CV vs. test. CV is much harsher than test (floor accuracy about 72% vs. 91%; mean error 12–14 m vs. 9 m). We ruled out location coverage: 9.9% of CV scans and 7.7% of test scans are more than 5 m from any training point. The likely cause is device novelty. In training, 17 of 18 users carried their own phone, so almost every CV validation scan comes from an unseen device, whereas 35.7% of test scans come from phones seen in training. User-grouped CV is therefore harder than the test set, but it is still a fair way to choose models.

The gap for Phase 2. All our models treat building, floor and position as separate tasks. The per-building forest links them only through a fixed cascade, and its two wrong-building scans are its worst errors. None of the models learns a representation of the 465 interacting APs that the three tasks could share.

VII. Conclusion and Phase 2 Plan

We built a leakage-free machine learning pipeline for Wi-Fi indoor positioning on UJIIndoorLoc. The main data decisions were to treat "not detected" as a very weak signal (−105 dBm) fitted on training data only, and to validate by holding out whole users. Simple raw features were as good as the engineered variants. A per-building Random Forest reaches 99.8% building accuracy, 91.3% floor accuracy and 9.37 m mean error on a test set collected months later with mostly unseen phones. k-NN is a close second. Remaining errors come from adjacent-floor confusion (especially building 1), scans that hear very few APs, and device differences.

In Phase 2 we will train a multitask MLP with a shared trunk and three heads (building and floor softmax, coordinate regression), optimised with $\mathcal{L} = \lambda_b \mathrm{CE}_b + \lambda_f \mathrm{CE}_f + \lambda_p \mathrm{MSE}_p$ [13], regularised with dropout, weight decay and early stopping on the same user-grouped validation. We will compare it with these baselines on the identical split and metrics, and run ablations (single-task vs. multitask, loss weights).

Team Contributions

Rishi Raj	Literature review, problem framing, evaluation metrics
Aditya Samadhiya	Data cleaning and exploratory data analysis
Yash Yadav	Feature engineering and feature comparison
Kaustubh Ranjan Mishra	Linear/logistic and k-NN models, cross-validation design
Yash Raj	Random Forest models, error analysis, repository and report

Reproducibility

Code, figures and an executed notebook are in the project GitHub repository. Running python scripts/download_data.py, run_eda.py and

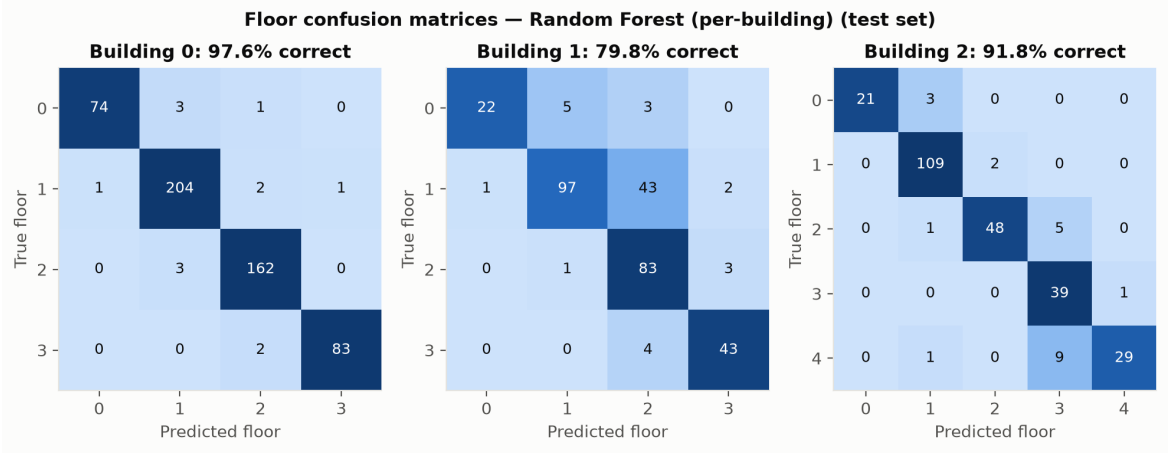


Fig. 12. Floor confusion matrices of the per-building Random Forest on the test set.



Fig. 13. Coordinate error map (per-building RF). Grey lines join the worst 10% of predictions to the true position.

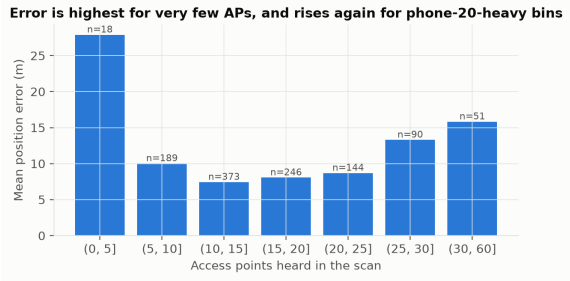


Fig. 14. Mean error by number of APs heard in the test scan.

run_experiments.py reproduces every number (random seed 42).

References

[1] F. Zafari, A. Gkelias, and K. K. Leung, "A survey of indoor localization systems and technologies," IEEE Communications Surveys & Tutorials, vol. 21, no. 3, pp. 2568–2599, 2019.

[2] P. Bahl and V. N. Padmanabhan, "RADAR: An in-building RF-based user location and tracking system," in Proc. IEEE INFOCOM, 2000, pp. 775–784.

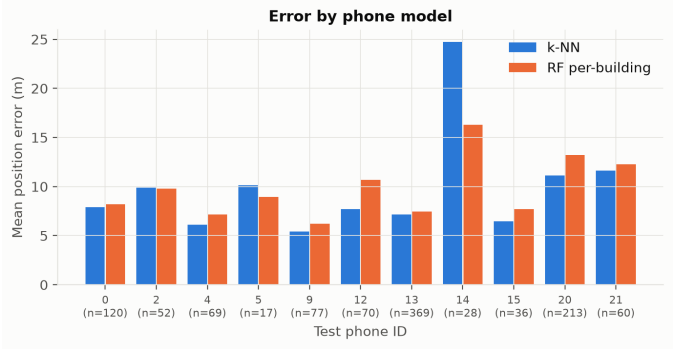


Fig. 15. Mean position error per test phone.

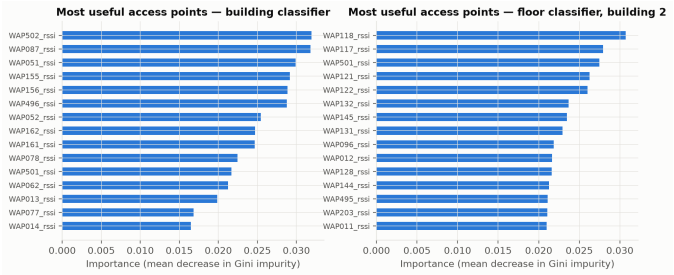


Fig. 16. Top-15 APs by Gini importance for the building and building-2 floor classifiers.

[3] M. Youssef and A. Agrawala, "The Horus WLAN location determination system," in Proc. ACM MobiSys, 2005, pp. 205–218.

[4] J. Torres-Sospedra, R. Montoliu, A. Martínez-Usó, J. P. Avariento, T. J. Arnau, M. Benedito-Bordonau, and J. Huerta, "UJIIndoorLoc: A new multi-building and multi-floor database for WLAN fingerprint-based indoor localization problems," in Proc. Int. Conf. Indoor Positioning and Indoor Navigation (IPIN), 2014, pp. 261–270.

[5] J. Torres-Sospedra, R. Montoliu, S. Trilles, Ó. Belmonte, and J. Huerta, "Comprehensive analysis of distance and similarity measures for Wi-Fi fingerprinting indoor positioning systems," Expert Systems with Applications, vol. 42, no. 23, pp. 9263–9278, 2015.

[6] J. Torres-Sospedra et al., "The smartphone-based offline indoor

location competition at IPIN 2016: Analysis and future work," Sensors, vol. 17, no. 3, p. 557, 2017.
[7] S. Bozkurt, G. Elibol, S. Gunal, and U. Yayan, "A comparative study on machine learning algorithms for indoor positioning," in Proc. Int. Symp. Innovations in Intelligent SysTems and Applications (INISTA), 2015.
[8] B. A. Akram, A. H. Akbar, and O. Shafiq, "HybLoc: Hybrid indoor Wi-Fi localization using soft clustering-based random decision forest ensembles," IEEE Access, vol. 6, pp. 38 251–38 272, 2018.
[9] L. Breiman, "Random forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.
[10] M. Nowicki and J. Wietrzykowski, "Low-effort place recognition with WiFi fingerprints using deep learning," in Automation 2017, Advances in Intelligent Systems and Computing, vol. 550. Springer, 2017.
[11] K. S. Kim, S. Lee, and K. Huang, "A scalable deep neural network architecture for multi-building and multi-floor indoor localization based on Wi-Fi fingerprinting," Big Data Analytics, vol. 3, p. 4, 2018.
[12] X. Song et al., "A novel convolutional neural network based indoor localization framework with WiFi fingerprinting," IEEE Access, vol. 7, 2019.
[13] R. Caruana, "Multitask learning," Machine Learning, vol. 28, no. 1, pp. 41–75, 1997.
[14] T. Cover and P. Hart, "Nearest neighbor pattern classification," IEEE Trans. Information Theory, vol. 13, no. 1, pp. 21–27, 1967.